

PADMA SESHADRI BALA BHAVAN SENIOR
SECONDARY SCHOOL, SIRUSERI (2024-25)

COMPUTER SCIENCE
BOARD PROJECT
STANDARD XII



SCHEDULEAZE

Automated School Time Table Maker

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XII B1

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BONAFIDE CERTIFICATE

Certified to the Bonafide record of work done by Master.
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P.G.T in:

Chennai

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Date:

Examiner:

Seal:

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Thanking you,

Yours Sincerely,

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SYNOPSIS

Objective

The primary aim of this project is to develop an automated system to create and manage school timetables using Python and MySQL. The system will optimize resource allocation (teachers, subjects, and classrooms) to generate error-free, conflict-free timetables while minimizing manual intervention.

Introduction

Scheduling school timetables manually can be a tedious, error-prone task, especially in schools with many classes, teachers, and subjects. This project aims to simplify and automate the process using a Python-based frontend and MySQL as the database management system. The tool will consider various constraints, such as subject periods, teacher availability, and simultaneous subjects, to ensure an efficient and balanced schedule.

Expected Outcome

The system will automate the creation of timetables, ensuring that no scheduling conflicts arise while saving time and reducing human errors. Schools will have a more efficient method for organizing their schedules, improving overall administrative workflow.

Conclusion

This project offers a practical solution to the complex problem of timetable creation by integrating Python's programming power with MySQL's robust database management, making it a valuable tool for schools of all sizes.

BACKEND STRUCTURE TABLE DESIGN

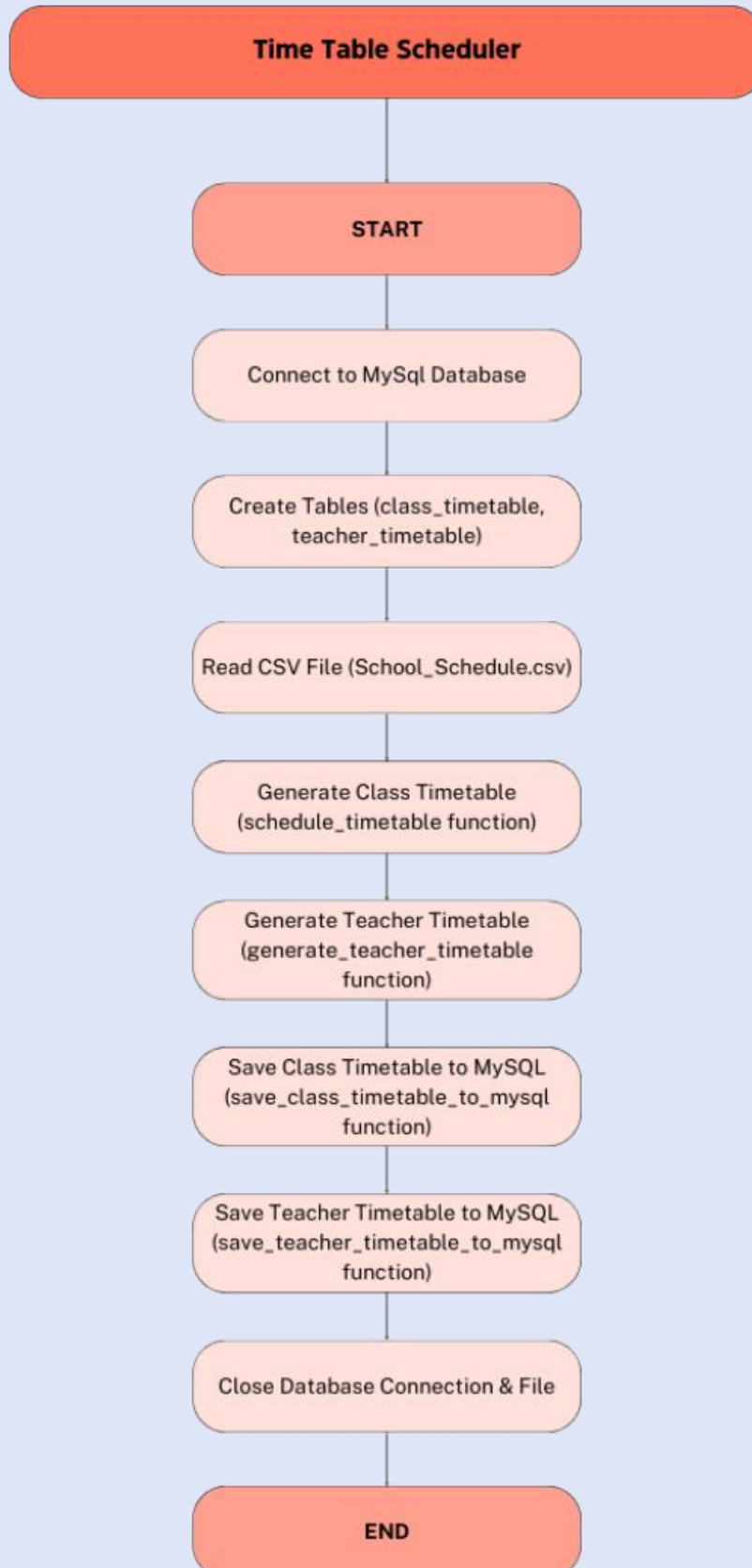
Class	Section	Subject	Teacher
XI	A	Mathematics	Deepa
XI	A	Physics	Karpagam
XI	A	Chemistry	Sampath
XI	A	English	Sudha
XI	A	Biology	Meena
XI	B	Mathematics	Deepa
XI	B	Physics	Karpagam
XI	B	Chemistry	Sampath
XI	B	English	Shinu
XI	B	Computer Science	Hemalatha
XII	A	Mathematics	Srividhya
XII	A	Physics	Karpagam
XII	A	Chemistry	Sampath
XII	A	English	Shinu
XII	A	Biology	Meena
XII	B	Mathematics	Deepa
XII	B	Physics	Karpagam
XII	B	Chemistry	Sampath
XII	B	English	Sudha
XII	B	Computer Science	Hemalatha

The backend structure is a csv file with
Class, Section, Subject, Teacher as columns.

ALGORITHM

1. 2 tables are created in the database school_db and the csv file with columns(Class, Section, Subject, Teacher) is opened in python.
2. A function is created to schedule time table as a dictionary with the given csv file, number of days and number of periods taken as user input.
3. A function is created for creating a teacher timetable as dictionary from the class timetable for which teachers have which class and subject at which time.
4. A function is created for adding data from the dictionary of class timetable to the table in mysql.
5. A function is created for adding data from the dictionary of teacher timetable to the table in mysql.
6. The functions are called out where the number of periods equals nine and the days are Monday, Tuesday, Wednesday, Thursday, Friday.

FLOW CHART



PROGRAM CODE

```
import csv
import random
import mysql.connector as mysql

con = mysql.connect(user = "root",host = "localhost",password = "India@2020", database = "school_db")
cur = con.cursor()
cur.execute("""CREATE TABLE IF NOT EXISTS class_timetable(
    class_section VARCHAR(50),
    day VARCHAR(20),
    period_1 VARCHAR(100),
    period_2 VARCHAR(100),
    period_3 VARCHAR(100),
    period_4 VARCHAR(100),
    period_5 VARCHAR(100),
    period_6 VARCHAR(100),
    period_7 VARCHAR(100),
    period_8 VARCHAR(100),
    period_9 VARCHAR(100),
    PRIMARY KEY (class_section, day));""")
cur.execute("""CREATE TABLE IF NOT EXISTS teacher_timetable(
    teacher VARCHAR(50),
    day VARCHAR(20),
    period_1 VARCHAR(100),
    period_2 VARCHAR(100),
    period_3 VARCHAR(100),
    period_4 VARCHAR(100),
    period_5 VARCHAR(100),
    period_6 VARCHAR(100),
    period_7 VARCHAR(100),
    period_8 VARCHAR(100),
    period_9 VARCHAR(100),
    PRIMARY KEY (teacher, day));""")

f = open("School_Schedule.csv","r")
read = csv.reader(f)
csvdata = []
for i in read:
    csvdata.append(i)

def schedule_timetable(data, days, periods):
    classes = []
    timetable = {}
    for i in range(1,len(data)):
        x = data[i][0] + "-" + data[i][1]
        if x not in classes:
            classes.append(x)

    for cls in classes:
        subjects = []
        teachers = []
        rows = {}
        for i in range(1,len(data)):
            if data[i][0] + "-" + data[i][1] == cls:
                subjects.append(data[i][2])
                teachers.append(data[i][3])

        for day in days:
            periodlst = []
            for period in range(periods):
                index = random.randint(0,len(subjects)-1)
                subject = subjects[index]
                teacher = teachers[index]
                subject_teach = (subject,teacher)
                periodlst.append(subject_teach)
            rows.update({day:periodlst})
        timetable[cls] = rows

    return timetable
```

```
def generate_teacher_timetable(timetable):
    teacher_timetable = {}
    for cls, days in timetable.items():
        for day, periods in days.items():
            for period_idx, entry in enumerate(periods):
                if entry: # If a subject and teacher is scheduled
                    subject, teacher = entry
                    if teacher not in teacher_timetable:
                        teacher_timetable[teacher] = {day: [None] * 9 for day in days}
                    teacher_timetable[teacher][day][period_idx] = f"{subject} ({cls})"
    return teacher_timetable
```

```
def save_class_timetable_to_mysql(connection, timetable):
    with connection.cursor() as cursor:
        for cls, days in timetable.items():
            for day, periods in days.items():
                formatted_periods = []
                for period in periods:
                    if period is None:
                        formatted_periods.append(None)
                    else:
                        subject, teacher = period
                        formatted_periods.append(f"{subject} ({teacher})")

    cursor.execute("""
        INSERT INTO class_timetable (class_section, day, period_1, period_2, period_3, period_4, period_5, period_6, period_7, period_8, period_9)
        VALUES (%s, %s, %s, %s, %s, %s, %s, %s, %s, %s, %s)
        ON DUPLICATE KEY UPDATE
        period_1 = VALUES(period_1), period_2 = VALUES(period_2), period_3 = VALUES(period_3),
        period_4 = VALUES(period_4), period_5 = VALUES(period_5), period_6 = VALUES(period_6),
        period_7 = VALUES(period_7), period_8 = VALUES(period_8), period_9 = VALUES(period_9);
    """, (cls, day, *formatted_periods))

    connection.commit()
```

```
def save_teacher_timetable_to_mysql(connection, timetable):
    with connection.cursor() as cursor:
        for teacher, days in timetable.items():
            for day, periods in days.items():
                cursor.execute("""
                    INSERT INTO teacher_timetable (teacher, day, period_1, period_2, period_3, period_4, period_5, period_6, period_7, period_8, period_9)
                    VALUES (%s, %s, %s, %s, %s, %s, %s, %s, %s, %s, %s)
                    ON DUPLICATE KEY UPDATE
                    period_1 = VALUES(period_1), period_2 = VALUES(period_2), period_3 = VALUES(period_3),
                    period_4 = VALUES(period_4), period_5 = VALUES(period_5), period_6 = VALUES(period_6),
                    period_7 = VALUES(period_7), period_8 = VALUES(period_8), period_9 = VALUES(period_9);
                """, (teacher, day, *(period if period else None for period in periods)))

    connection.commit()
```

```
time_table = schedule_timetable(csvdata, ["Monday", "Tuesday", "Wednesday"], 9)
teacher_time_table = generate_teacher_timetable(time_table)
save_class_timetable_to_mysql(con, time_table)
save_teacher_timetable_to_mysql(con, teacher_time_table)
print("Time table saved to SQL!")
con.close()
f.close()
```

SAMPLE OUTPUT

In Python:

```
Python 3.10.0 (tags/v3.10.0:b494f59, Oct 4 2021, 19:00:18) [MSC v.1929 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
== RESTART: D:\Madhu\Python\Investigatory Project XII\Time Table Scheduler.py ==
Time table saved to SQL!
>>>
```

In MySql:

Database has 2 tables

	Tables_in_school_db
►	class_timetable
	teacher_timetable

Class Timetable

class_section	day	period_1	period_2	period_3	period_4	period_5	period_6	period_7	period_8	period_9
XI-A	1Monday	Biology (Meena)	Physics (Karpagam)	English (Sudha)	Chemistry (Sampath)	Mathematics (Deepa)	English (Sudha)	Biology (Meena)	Physics (Karpagam)	Biology (Meena)
XI-A	2Tuesday	Chemistry (Sampath)	Biology (Meena)	Mathematics (Deepa)	Chemistry (Sampath)	Chemistry (Sampath)	English (Sudha)	Chemistry (Sampath)	Mathematics (Deepa)	Mathematics (Deepa)
XI-A	3Wednesday	English (Sudha)	Mathematics (Deepa)	Physics (Karpagam)	Physics (Karpagam)	Chemistry (Sampath)	Chemistry (Sampath)	Mathematics (Deepa)	Physics (Karpagam)	Physics (Karpagam)
XI-A	4Thursday	Chemistry (Sampath)	Chemistry (Sampath)	Biology (Meena)	Biology (Meena)	Biology (Meena)	Mathematics (Deepa)	Biology (Meena)	English (Sudha)	Biology (Meena)
XI-A	5Friday	Mathematics (Deepa)	Physics (Karpagam)	English (Sudha)	Biology (Meena)	Biology (Meena)	English (Sudha)	Biology (Meena)	Chemistry (Sampath)	Chemistry (Sampath)
XI-B	1Monday	Computer Science (Hemalatha)	Computer Science (Hemalatha)	Mathematics (Deepa)	Physics (Karpagam)	Chemistry (Sampath)	Mathematics (Deepa)	Chemistry (Sampath)	Physics (Karpagam)	Physics (Karpagam)
XI-B	2Tuesday	Physics (Karpagam)	Computer Science (Hemalatha)	Mathematics (Deepa)	English (Shinu)	Chemistry (Sampath)	Computer Science (Hemalatha)	Physics (Karpagam)	Computer Science (Hemalatha)	Mathematics (Deepa)
XI-B	3Wednesday	Computer Science (Hemalatha)	Physics (Karpagam)	Mathematics (Deepa)	Computer Science (Hemalatha)	Physics (Karpagam)	Physics (Karpagam)	Chemistry (Sampath)	Mathematics (Deepa)	Computer Science (Hemalatha)
XI-B	4Thursday	Mathematics (Deepa)	Physics (Karpagam)	Computer Science (Hemalatha)	Chemistry (Sampath)	English (Shinu)	Chemistry (Sampath)	Physics (Karpagam)	Physics (Karpagam)	Physics (Karpagam)
XI-B	5Friday	Mathematics (Deepa)	Mathematics (Deepa)	Physics (Karpagam)	English (Shinu)	Computer Science (Hemalatha)	Computer Science (Hemalatha)	Chemistry (Sampath)	Computer Science (Hemalatha)	Mathematics (Deepa)
XII-A	1Monday	English (Shinu)	Biology (Meena)	Biology (Meena)	Chemistry (Sampath)	Chemistry (Sampath)	English (Shinu)	Chemistry (Sampath)	Physics (Karpagam)	Chemistry (Sampath)
XII-A	2Tuesday	Mathematics (Srividhya)	Physics (Karpagam)	Biology (Meena)	Mathematics (Srividhya)	Biology (Meena)	English (Shinu)	Mathematics (Srividhya)	Mathematics (Srividhya)	English (Shinu)
XII-A	3Wednesday	Mathematics (Srividhya)	Mathematics (Srividhya)	Biology (Meena)	English (Shinu)	Chemistry (Sampath)	Chemistry (Sampath)	Physics (Karpagam)	Mathematics (Srividhya)	Chemistry (Sampath)
XII-A	4Thursday	Mathematics (Srividhya)	English (Shinu)	Biology (Meena)	Biology (Meena)	Mathematics (Srividhya)	English (Shinu)	English (Shinu)	Mathematics (Srividhya)	Biology (Meena)
XII-A	5Friday	Mathematics (Srividhya)	English (Shinu)	Physics (Karpagam)	English (Shinu)	Biology (Meena)	English (Shinu)	Chemistry (Sampath)	Physics (Karpagam)	Physics (Karpagam)
XII-B	1Monday	Computer Science (Hemalatha)	Mathematics (Deepa)	Mathematics (Deepa)	Mathematics (Deepa)	Physics (Karpagam)	Chemistry (Sampath)	Physics (Karpagam)	Chemistry (Sampath)	Mathematics (Deepa)
XII-B	2Tuesday	Physics (Karpagam)	Chemistry (Sampath)	English (Sudha)	English (Sudha)	English (Sudha)	English (Sudha)	Chemistry (Sampath)	Mathematics (Deepa)	Mathematics (Deepa)
XII-B	3Wednesday	English (Sudha)	Chemistry (Sampath)	Mathematics (Deepa)	Mathematics (Deepa)	Computer Science (Hemalatha)	Physics (Karpagam)	Mathematics (Deepa)	Chemistry (Sampath)	Physics (Karpagam)
XII-B	4Thursday	Computer Science (Hemalatha)	English (Sudha)	Chemistry (Sampath)	Physics (Karpagam)	Mathematics (Deepa)	Mathematics (Deepa)	Chemistry (Sampath)	Computer Science (Hemalatha)	English (Sudha)
XII-B	5Friday	Computer Science (Hemalatha)	Mathematics (Deepa)	Computer Science (Hemalatha)	Chemistry (Sampath)	Mathematics (Deepa)	Chemistry (Sampath)	Computer Science (Hemalatha)	Chemistry (Sampath)	Mathematics (Deepa)

Teacher Timetable

	teacher	day	period_1	period_2	period_3	period_4	period_5	period_6	period_7	period_8	period_9
▶	Deepa	1Monday	NONE	Mathematics (XII-B)	Mathematics (XII-B)	Mathematics (XII-B)	Mathematics (XI-A)	Mathematics (XI-B)	NONE	NONE	Mathematics (XII-B)
	Deepa	2Tuesday	NONE	NONE	Mathematics (XI-B)	NONE	NONE	NONE	NONE	NONE	Mathematics (XII-B)
	Deepa	3Wednesday	NONE	Mathematics (XI-A)	Mathematics (XII-B)	Mathematics (XII-B)	NONE	NONE	Mathematics (XII-B)	Mathematics (XI-B)	NONE
	Deepa	4Thursday	Mathematics (XI-B)	NONE	NONE	NONE	Mathematics (XII-B)	Mathematics (XII-B)	NONE	NONE	NONE
	Deepa	5Friday	Mathematics (XI-B)	Mathematics (XII-B)	NONE	NONE	Mathematics (XII-B)	NONE	NONE	NONE	Mathematics (XII-B)
	Hemalatha	1Monday	Computer Science (XII-B)	Computer Science (XI-B)	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	Hemalatha	2Tuesday	NONE	Computer Science (XI-B)	NONE	NONE	NONE	Computer Science (XI-B)	NONE	Computer Science (XI-B)	NONE
	Hemalatha	3Wednesday	Computer Science (XI-B)	NONE	NONE	Computer Science (XI-B)	Computer Science (XII-B)	NONE	NONE	NONE	Computer Science (XI-B)
	Hemalatha	4Thursday	Computer Science (XII-B)	NONE	Computer Science (XI-B)	Computer Science (XI-B)	NONE	NONE	Computer Science (XII-B)	Computer Science (XI-B)	NONE
	Hemalatha	5Friday	Computer Science (XII-B)	NONE	Computer Science (XII-B)	NONE	Computer Science (XI-B)	Computer Science (XI-B)	Computer Science (XII-B)	Computer Science (XI-B)	NONE
	Karpagam	1Monday	NONE	Physics (XI-A)	NONE	Physics (XI-B)	Physics (XII-B)	NONE	Physics (XII-B)	Physics (XII-A)	Physics (XI-B)
	Karpagam	2Tuesday	Physics (XII-B)	Physics (XII-A)	NONE	NONE	NONE	Physics (XII-B)	Physics (XI-B)	NONE	NONE
	Karpagam	3Wednesday	NONE	Physics (XI-A)	Physics (XI-A)	Physics (XI-A)	Physics (XI-B)	Physics (XII-B)	Physics (XI-A)	Physics (XI-A)	Physics (XII-B)
	Karpagam	4Thursday	NONE	Physics (XI-B)	NONE	Physics (XII-B)	NONE	NONE	Physics (XII-A)	Physics (XI-B)	Physics (XII-A)
	Karpagam	5Friday	NONE	Physics (XI-A)	Physics (XII-A)	NONE	NONE	NONE	NONE	Physics (XI-B)	Physics (XII-A)
	Meena	1Monday	Biology (XI-A)	Biology (XII-A)	Biology (XII-A)	NONE	NONE	NONE	Biology (XI-A)	NONE	Biology (XI-A)
	Meena	2Tuesday	NONE	Biology (XII-A)	Biology (XII-A)	NONE	Biology (XII-A)	NONE	NONE	NONE	NONE
	Meena	3Wednesday	NONE	NONE	Biology (XII-A)	NONE	NONE	NONE	NONE	NONE	NONE
	Meena	4Thursday	NONE	NONE	Biology (XII-A)	Biology (XII-A)	Biology (XI-A)	NONE	Biology (XI-A)	NONE	Biology (XII-A)
	Meena	5Friday	NONE	NONE	NONE	Biology (XI-A)	Biology (XII-A)	NONE	NONE	Biology (XI-A)	NONE
	Sampath	1Monday	NONE	NONE	NONE	Chemistry (XII-A)	Chemistry (XII-A)	Chemistry (XII-B)	Chemistry (XII-A)	Chemistry (XII-B)	Chemistry (XII-A)

When we want a specific class

	class_section	day	period_1	period_2	period_3	period_4	period_5	period_6	period_7	period_8	period_9
▶	XII-B	1Monday	Computer Science (Hemalatha)	Mathematics (Deepa)	Mathematics (Deepa)	Mathematics (Deepa)	Physics (Karpagam)	Chemistry (Sampath)	Physics (Karpagam)	Chemistry (Sampath)	Mathematics (Deepa)
	XII-B	2Tuesday	Physics (Karpagam)	Chemistry (Sampath)	English (Sudha)	English (Sudha)	English (Sudha)	Physics (Karpagam)	English (Sudha)	Chemistry (Sampath)	Mathematics (Deepa)
	XII-B	3Wednesday	English (Sudha)	Chemistry (Sampath)	Mathematics (Deepa)	Mathematics (Deepa)	Computer Science (Hemalatha)	Physics (Karpagam)	Mathematics (Deepa)	Chemistry (Sampath)	Physics (Karpagam)
	XII-B	4Thursday	Computer Science (Hemalatha)	English (Sudha)	Chemistry (Sampath)	Physics (Karpagam)	Mathematics (Deepa)	Mathematics (Deepa)	Chemistry (Sampath)	Computer Science (Hemalatha)	English (Sudha)
	XII-B	5Friday	Computer Science (Hemalatha)	Mathematics (Deepa)	Computer Science (Hemalatha)	Chemistry (Sampath)	Mathematics (Deepa)	Chemistry (Sampath)	Computer Science (Hemalatha)	Chemistry (Sampath)	Mathematics (Deepa)

When we want a specific teacher's time table

	teacher	day	period_1	period_2	period_3	period_4	period_5	period_6	period_7	period_8	period_9
▶	Hemalatha	1Monday	Computer Science (XII-B)	Computer Science (XI-B)	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	Hemalatha	2Tuesday	NONE	Computer Science (XI-B)	NONE	NONE	NONE	Computer Science (XI-B)	NONE	Computer Science (XI-B)	NONE
	Hemalatha	3Wednesday	Computer Science (XI-B)	NONE	NONE	Computer Science (XI-B)	Computer Science (XII-B)	NONE	NONE	NONE	Computer Science (XI-B)
	Hemalatha	4Thursday	Computer Science (XII-B)	NONE	Computer Science (XI-B)	Computer Science (XI-B)	NONE	NONE	Computer Science (XII-B)	Computer Science (XI-B)	NONE
	Hemalatha	5Friday	Computer Science (XII-B)	NONE	Computer Science (XII-B)	NONE	Computer Science (XI-B)	Computer Science (XI-B)	Computer Science (XII-B)	Computer Science (XI-B)	NONE

LIMITATIONS OF THE PROJECT

1. Subjects will be repeated for the same class on same day.
2. Subjects and teachers are assigned randomly, leading to possible imbalances or unrealistic schedules. The project doesn't distribute subjects or teachers evenly across days and periods.
3. There's no mechanism to handle scheduling conflicts, such as a teacher being assigned to multiple classes simultaneously.
4. It doesn't consider teacher availability or constraints

SCOPE FOR IMPROVEMENT

1. Implement logic to check for and resolve conflicts, ensuring a teacher or subject is not double-booked.
2. Add validation for CSV data (e.g., check for missing columns, invalid entries).
3. Use more sophisticated algorithms (e.g., constraint satisfaction, optimization techniques) for fair and balanced scheduling.
4. Develop a graphical user interface (GUI) or web application for easier interaction.
5. Add functionality to export timetables in formats like PDF, Excel, or web pages.
6. Add notifications (email/SMS) for schedule updates or conflicts.

BIBLIOGRAPHY

<https://www.w3schools.com/python/>

<https://www.geeksforgeeks.org/>

<https://www.mysql.com/>